

Exam examples

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1 Problem 1

We have a differential drive robot with wheels encoder (odometry). Assume it moves on the plane but the surface is not planar.

The robot has a sensor to perceive 3D landmark (not places at the center). Model an estimation problem that is able to estimate the landmark position assumed no data association.

Locally, the robot assumes it's moving on a plane; we assume the robot can only move on the spot. having defined

$$\Delta\rho = \frac{\Delta_r k_r}{2} + \frac{\Delta_l k_l}{2} \quad (1.1)$$

The transformation is

$$\begin{aligned} \Delta x &\approx \Delta\rho \\ \Delta y &= 0 \\ \Delta\theta &= \frac{\Delta_r k_r - \Delta_l k_l}{b} \end{aligned} \quad (1.2)$$

Note: the dimension of the wheels r is embedded in Δ_r and Δ_l ; in particular it depends on the dimension of the wheel:

$$\begin{aligned} \Delta_r k_r &= \left(R + \frac{b}{2}\right) \Delta\theta \\ \Delta_l k_l &= \left(R - \frac{b}{2}\right) \Delta\theta \end{aligned} \quad (1.3)$$

The domain is

- $k_l, k_r, b \in \mathbb{R}^3$,
- $\Delta_l, \Delta_r \in \mathbb{R}^2$.

Defined the function x^{odom} as $x^{odom} : k_l, k_r, b, \Delta_l, \Delta_r \rightarrow SE(2)$

Since we want to talk in x (for the states), we define

- $x^{odom} \in \mathbb{R}^3$,
- $x_l \triangleq k_l$,
- $x_r \triangleq k_r$,
- $x_b \triangleq b$.

The calibration parameter are becoming part of the estimation problem, so part of the state:

$$h_{odom_i} : x^{odom} \rightarrow SE(2) \quad (1.4)$$

What do we have in the state for this SLAM problem:

- poses of the robot: $x_{robot} \in SE(3) \times N$ (we are estimating a trajectory),
- the odometry parameters: $x_{odom} \in \mathbb{R}^3$,
- positions of the M landmarks: $x_{landmarks} \in \mathbb{R}^3 \times M$,

¹The numbers of tics Δ_r

²Negligible, so assumed to be 0

³ b is the baseline, so the distance between the two wheels

- pose of the sensor (since we don't know where it is): $x_{sensors} \in SE(3)$

The perturbation vector is in figure 1.1:

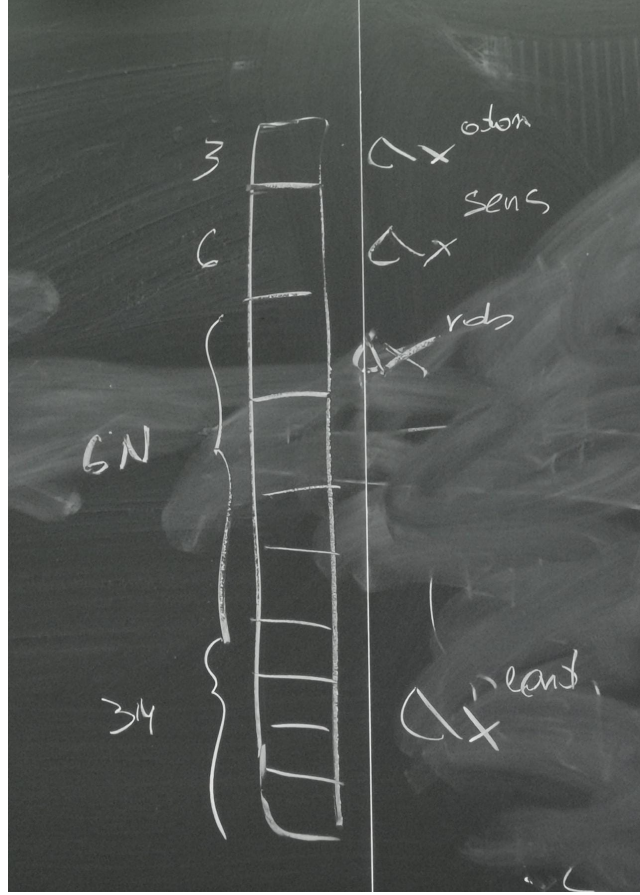


Figure 1.1: Perturbation vector

Let's move to the measurement (we consider one measurement at the time):

$$Z_{land} \in R^3 \quad (1.5)$$

in the measurement we have the tick, how to get them? We can predict them starting from h_{odom} . Remember that

$$\Delta x = \frac{k_r \Delta_r + k_l \Delta_l}{2} = k_r \Delta_r + k_l \Delta_l \quad ^4 \quad (1.6)$$

$$b \Delta \theta = k_r \Delta_r - k_l \Delta_l \quad (1.7)$$

by summing and subtracting equations 1.6 and 1.7, we can have an expression of the ticks:

$$\begin{aligned} \Delta x + b \Delta \theta &= 2k_r \Delta_r \\ \Delta x - b \Delta \theta &= 2k_l \Delta_l \end{aligned} \quad (1.8)$$

From this we get a function that depends on the calibration and odometry parameters

$$g(\Delta x, \Delta \theta) \rightarrow \Delta_r, \Delta_l \quad (1.9)$$

where

$$\begin{aligned} \Delta_r &= \frac{\Delta x + b \Delta \theta}{2k_r} \\ \Delta_l &= \frac{\Delta x - b \Delta \theta}{2k_l} \end{aligned} \quad (1.10)$$

⁴We absorbed the 2 in the constants k_r and k_l

We have a function that, given an angle and the translation, we are measuring the encoders. So we have the observation of the odometry:

$$Z_{odom} \in R^2 \quad (1.11)$$

R^2 since we are considering Z_l and Z_r , so the odometry for both wheels.

To predict the odometry measurement we need:

- x_{rob_i} , current position of the robot,
- $x_{rob_{i+1}}$, immediate next position of the robot,
- x_{odom} , odometry parameter.

What is the structure of the Jacobian of one of the two functions (so of the measurement)? Is tall 2 (the number of measurement) and wide 3 (the number of states that influence our measurement, so x_{rob_i} , $x_{rob_{i+1}}$ and x_{odom}).

Let's now determine the prediction h , we are not on a manifold so no *boxminus*/*boxplus* operators.

We need to populate equation 1.9, to do so we need the numbers: we need $\Delta\theta$ and Δx

We need to define the relative motion dx/dt from current position to next one: $(X_i^{-1}X_{i+1})$ this is basically $R|t$.

From this we can extract the rotation matrix after the motion

$$R = \begin{pmatrix} c_\theta & s_\theta & 0 \\ s_\theta & -c_\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (1.12)$$

Just like Robotics 1 taught us (almeno questo lo ricordo, yeee), we can easily determine $\Delta\theta$:

$$\Delta\theta = \text{atan2}(s_\theta, c_\theta) \quad (1.13)$$

Δx can be easily determined from the translation vector t .

Let's now measure the sensor:

$$h[n, m] = x_{sens}^{-1} x_{rob}^{-1} x_{land} \quad (1.14)$$

We can now do the structure of the Jacobian: tall 3 and as wide as x_{rob_n} , x_{land_n} and x_{sens} . This Jacobian will contribute with 9 non zero blocks to the H matrix (since we will consider the Jacobian times its transpose, so $3 \times 3 = 9$). What if we add another sensor? The states stays the same, you just have to add something to this Jacobian and to the H matrix.

2 Projects

You get some data, work on your stuff, push it to the the TAs, they run it and test it with different data.